ASIC Design Flow Tutorial

Lab 2a: Logic Synthesis. Design Compiler

A. Laboratory tasks

Introduction

Using Design Compiler, it is possible to:

- Produce fast, area-efficient ASIC designs by employing user-specified or standard-cell libraries
- Translate designs from one technology to another
- Explore design tradeoffs involving design constraints such as timing, area, and power under various loading, temperature, and voltage conditions
- Synthesize and optimize finite state machines
- Integrate netlist inputs and netlist or schematic outputs into third-party environments while still supporting delay information and place and route constraints
- Create and partition hierarchical schematics automatically.

Basic Synthesis Flow

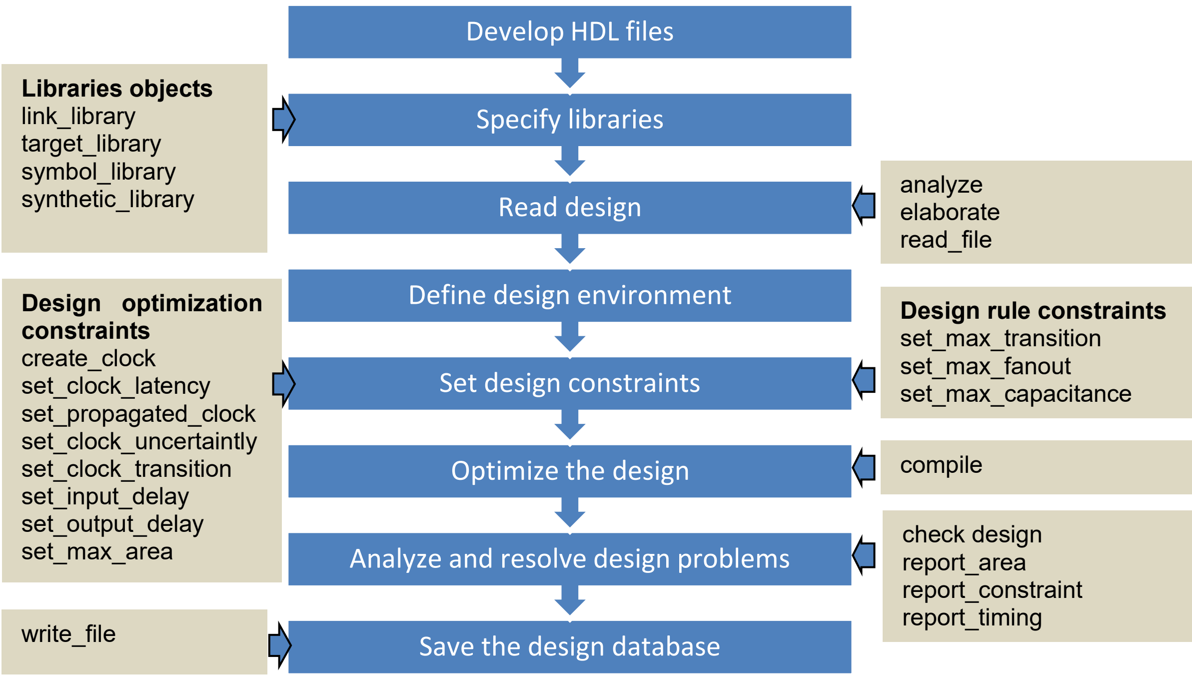
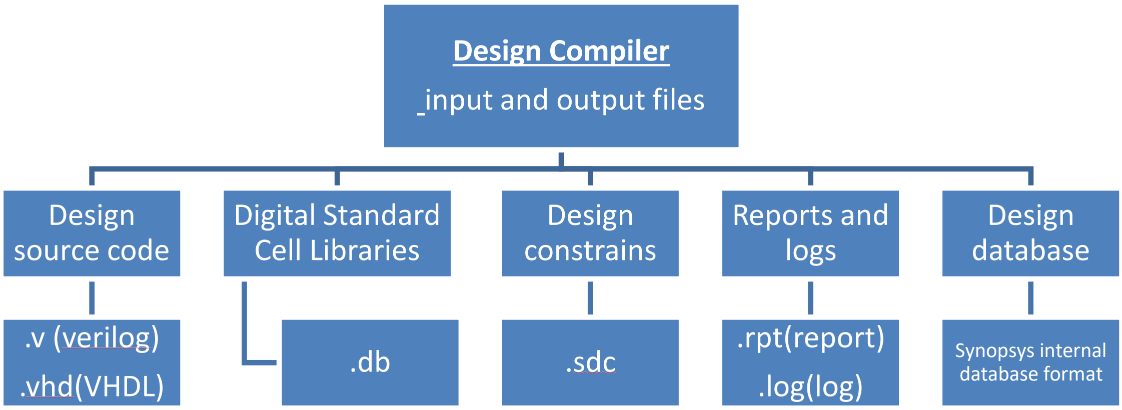


Fig.1.Basic synthesis flow

Design Compiler input and output files



Note: source /home/share/Env/synopsys_setup.env to open all tools.

1. Start Design Compiler graphical user interface (GUI) from the **~/TH_TKVM_CLC/Digital_Synopsys/syn** directory. To start it use the following command:

```
% dc_shell
dc_shell> start_gui
```

This opens the Design Compiler top-level GUI window (Design Vision). (Fig. 1)

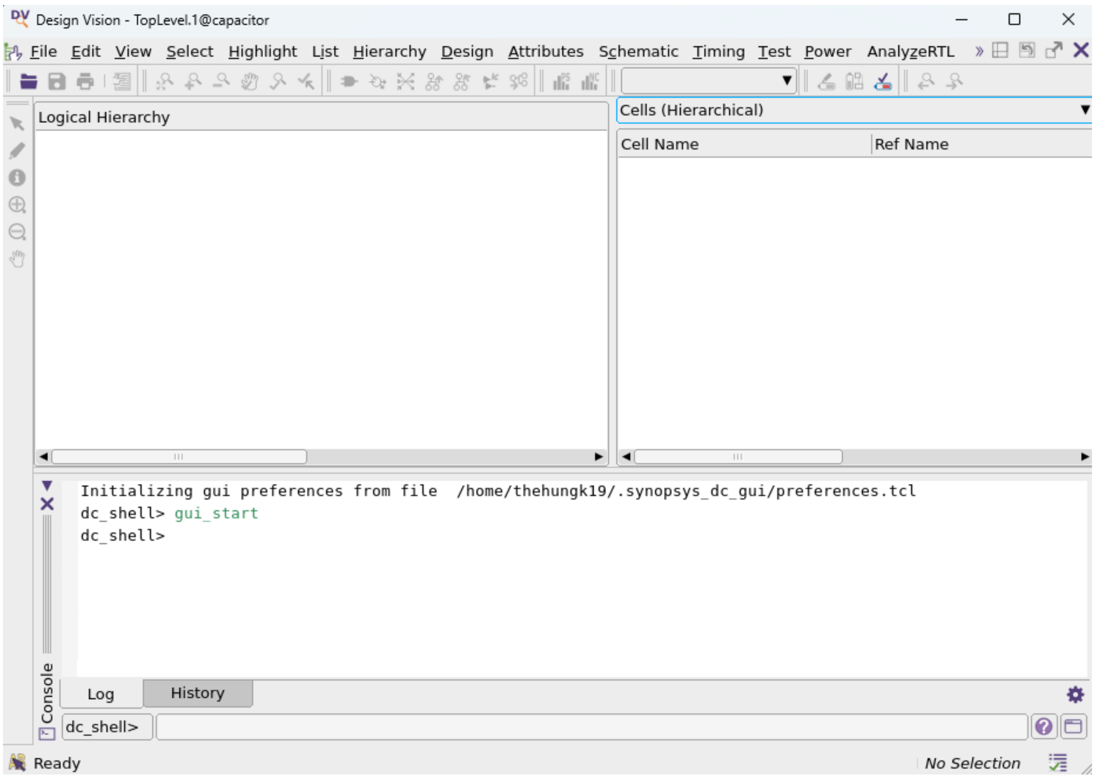


Fig.1. Design Compiler Top-level GUI window

2. Put your cursor in the terminal to type commands (Fig. 2).

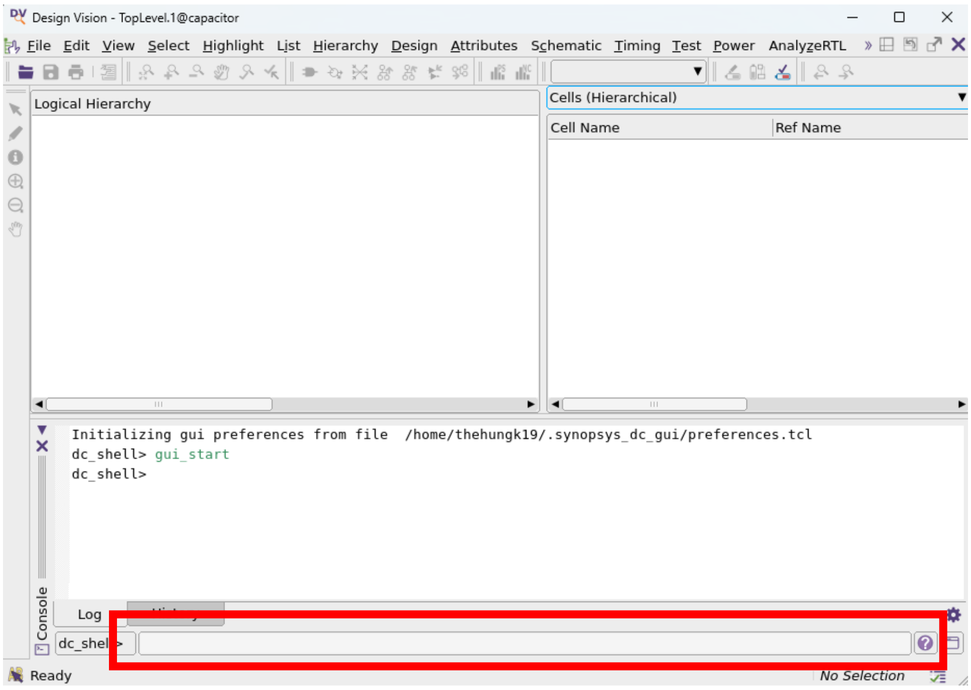


Fig.2. Application setup window

3. Set up the initial design:

```
dc_shell> set top_module johnson
dc_shell> set_svf $top_module.svf
dc_shell> define_design_lib work -path ./work
```

4. Define libraries for the process of synthesis:

```
dc_shell> lappend search_path ../Lib/db
dc_shell> lappend search_path ../cons
dc_shell> lappend search_path ../RTL
dc_shell> set LIB "saed32rvt_tt1p05v25c.db"
dc_shell> set target_library [list $LIB]
dc_shell> set link_library [list * $LIB]
```

5. Read and compile the syntax of Verilog files that are from RTL coding:

```
dc_shell> set file_format verilog
dc_shell> analyze -format $file_format {johnson.v}
dc_shell> elaborate -lib work johnson
dc_shell> current_design $top_module
```

6. Link Verilog files and Libs for checking pre-synthesis:

```
dc_shell> link
dc_shell> check_design
```

```
dc_shell> link
  Linking design 'johnson'
  Using the following designs and libraries:
  --------------------------------------------------------------------------
  johnson                    /home/thehungk19/teaching_assistant/Synopsys_Practice/TH_TKVM_CLC/Digital_Synopsys/syn/johnson.db
  saed32rvt_tt1p05v25c (library) /home/thehungk19/teaching_assistant/Synopsys_Practice/TH_TKVM_CLC/Digital_Synopsys/Lib/db/saed32rvt_tt1p05v25c.db

1
dc_shell> check_design
1
```

Fig.3. Result for linking and checking pre-synthesis.

7. Read Synopsys Design Constraints (.sdc) and set up timing analysis:

```
dc_shell> group_path -name INREG -from [all_inputs]
dc_shell> group_path -name REGOUT -to [all_outputs]
dc_shell> group_path -name INOUT -from [all_inputs] -to [all_outputs]
dc_shell> source -echo dc.sdc
```

8. Start synthesis by using the following commands:

```
dc_shell> set case_analysis_with_logic_constants true
dc_shell> set_fix_multiple_port_nets -all
dc_shell> compile_ultra -no_autoungroup
dc_shell> set_svf -off
```

9. Analyze reports about timing, area, and power:

```
dc_shell> report_area -hierarchy
```

```
Number of ports:                          10
Number of nets:                           12
Number of cells:                           9
Number of combinational cells:             1
Number of sequential cells:                8
Number of macros/black boxes:              0
Number of buf/inv:                         1
Number of references:                      2

Combinational area:                 1.524864
Buf/Inv area:                       1.524864
Noncombinational area:             56.928257
Macro/Black Box area:               0.000000
Net Interconnect area:              3.448122

Total cell area:                   58.453121
Total area:                        61.901243
```

```
dc_shell> report_timing -max_paths 100 -delay_type min
```

```
Startpoint: out_reg[5] (rising edge-triggered flip-flop clocked by clk)
Endpoint: out_reg[6] (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
johnson            ForQA                 saed32rvt_tt1p05v25c

Point                                    Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                    0.00       0.00
clock network delay (ideal)              0.00       0.00
out_reg[5]/CLK (DFFARX1_RVT)             0.00       0.00 r
out_reg[5]/Q (DFFARX1_RVT)               0.12       0.12 f
out_reg[6]/D (DFFARX1_RVT)               0.01       0.13 f
data arrival time                                   0.13

clock clk (rise edge)                    0.00       0.00
clock network delay (ideal)              0.00       0.00
clock uncertainty                        0.10       0.10
out_reg[6]/CLK (DFFARX1_RVT)             0.00       0.10 r
library hold time                        0.00       0.10
data required time                                  0.10
-----------------------------------------------------------
data required time                                  0.10
data arrival time                                  -0.13
-----------------------------------------------------------
slack (MET)                                         0.03
```

```
dc_shell> report_timing -max_paths 100 -delay_type max
```

```
Startpoint: out_reg[7] (rising edge-triggered flip-flop clocked by clk)
Endpoint: out_reg[0] (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
johnson            ForQA                 saed32rvt_tt1p05v25c

Point                                    Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                    0.00       0.00
clock network delay (ideal)              0.00       0.00
out_reg[7]/CLK (DFFARX1_RVT)             0.00       0.00 r
out_reg[7]/QN (DFFARX1_RVT)              0.09       0.09 r
out_reg[0]/D (DFFARX1_RVT)               0.01       0.10 r
data arrival time                                   0.10

clock clk (rise edge)                  250.00     250.00
clock network delay (ideal)              0.00     250.00
clock uncertainty                       -0.25     249.75
out_reg[0]/CLK (DFFARX1_RVT)             0.00     249.75 r
library setup time                      -0.03     249.72
data required time                                249.72
-----------------------------------------------------------
data required time                                249.72
data arrival time                                  -0.10
-----------------------------------------------------------
slack (MET)                                       249.62
```

```
dc_shell> report_qor
```

```
Hostname: capacitor

Compile CPU Statistics
-----------------------------------------
Resource Sharing:                    0.00
Logic Optimization:                  0.04
Mapping Optimization:                0.25
-----------------------------------------
Overall Compile Time:                3.89
Overall Compile Wall Clock Time:     4.24

--------------------------------------------------------------------

Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0


Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

--------------------------------------------------------------------
```

9.9 Write important output files for the next steps (.v, .ddc, .sdc, …)

```
dc_shell> set verilogout_equation  false

dc_shell> write_file -format verilog -hierarchy -output
./output/design_mapped.v

dc_shell> write_file -format ddc -hierarchy -output
./output/design_mapped.ddc

dc_shell> write_sdc  -nosplit ../cons/icc2.sdc

dc_shell> write_sdf           ./output/design_mapped.sdf
```

10. To exit Design Compiler write exit in the command line.

```
dc_shell> exit
```

B. Homework

Synthesize 2 designs in the Lab 1, including the combinational circuit and the 8-bit counter circuit.